

Degree & Branch	B.E. Computer Science & Engineering	Semester	V
Subject Code & Name	ICS1512 & Machine Learning Algorithms Laboratory		
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Experiment 2: Loan Amount Prediction using Linear Regression

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1. Aim

To apply Linear Regression to predict the loan amount sanctioned to users using a dataset of historical loan records and user features.

2. Libraries Used

- **Pandas:** For data loading, cleaning, and manipulation.
- **NumPy:** For numerical operations, especially for performance metrics.
- **Matplotlib & Seaborn:** For data visualization, including histograms, scatter plots, and heatmaps.
- **Scikit-learn:** For splitting data, model creation (LinearRegression), preprocessing (StandardScaler), and evaluation.

3. Objective

- Build and evaluate a Linear Regression model to predict loan amounts.
- Perform data preprocessing, Exploratory Data Analysis (EDA), and feature engineering.
- Visualize important relationships within the data and the model's results.
- Measure model performance using standard regression metrics like MAE, MSE, RMSE, and R^2 .

4. Mathematical Description

The core objective of Linear Regression is to find the best-fitting line through the data points by minimizing the sum of the squared differences between the actual values (y_i) and the predicted values ($\hat{y}_i$). This is achieved by minimizing the Mean Squared Error (MSE).

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

5. Code Summary

The following Python code summarizes the key steps from data preparation to model evaluation.

```
1 import pandas as pd
2 import numpy as np
3 from sklearn.model_selection import train_test_split
4 from sklearn.linear_model import LinearRegression
5 from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
6
7 # 1. Load and preprocess data (handle missing values, encode categoricals)
8 df = pd.read_csv('train.csv')
9 # ... (Median/mode imputation and one-hot encoding as in script) ...
10 df_fe_encoded = pd.get_dummies(df_fe) # Assuming df_fe is feature engineered df
11
12 # 2. Define features (X) and target (y)
13 target = 'Loan Amount Request (USD)'
14 X = df_fe_encoded.drop(columns=[target])
15 y = df_fe_encoded[target]
16
17 # 3. Split data into training, validation, and test sets (60:20:20)
18 X_trainval, X_test, y_trainval, y_test = train_test_split(X, y, test_size=0.2,
19     random_state=42)
20
21 X_train, X_val, y_train, y_val = train_test_split(X_trainval, y_trainval,
22     test_size=0.25, random_state=42)
23
24 # 4. Train the Linear Regression model
25 model = LinearRegression()
26 model.fit(X_train, y_train)
27
28 # 5. Make predictions on the test set
29 y_test_pred = model.predict(X_test)
30
31 # 6. Evaluate the model performance
32 mae_test = mean_absolute_error(y_test, y_test_pred)
33 mse_test = mean_squared_error(y_test, y_test_pred)
34 rmse_test = np.sqrt(mse_test)
35 r2_test = r2_score(y_test, y_test_pred)
36
37 print(f"Test MAE: {mae_test:.2f}")
38 print(f"Test RMSE: {rmse_test:.2f}")
39 print(f"Test R2 Score: {r2_test:.4f}")
```

Listing 1: Key steps for Linear Regression modeling.

6. Included Plots

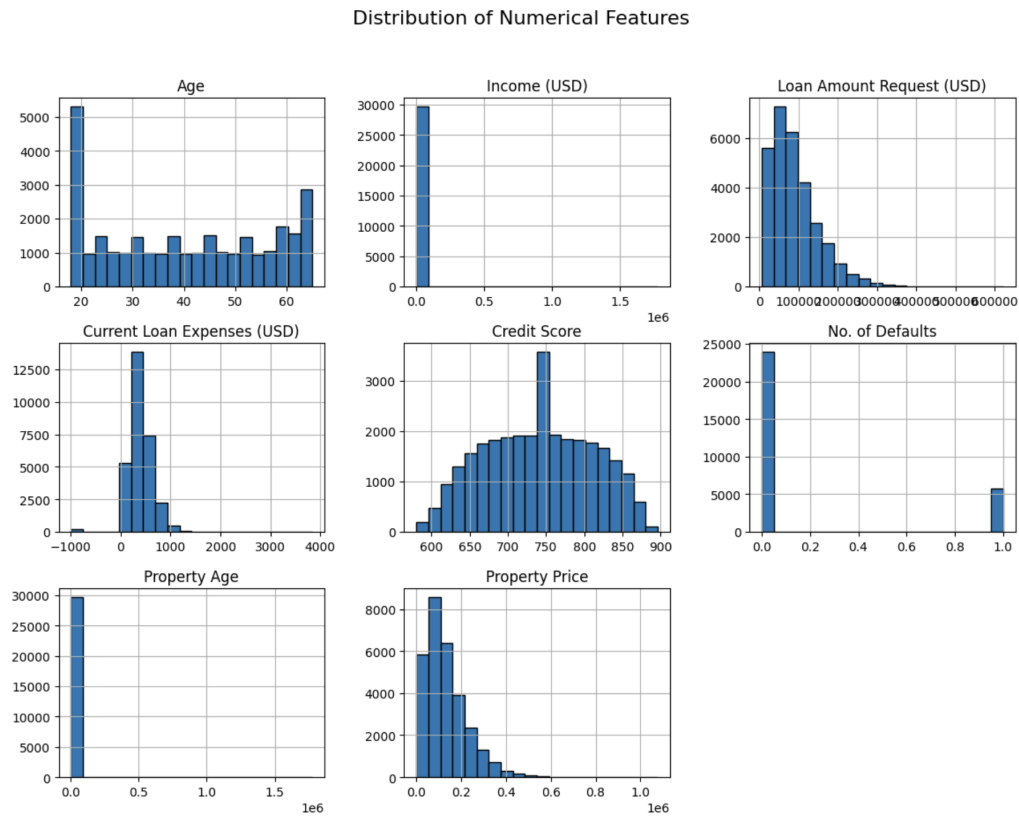


Figure 1: Distribution of Loan Amount Request (USD).

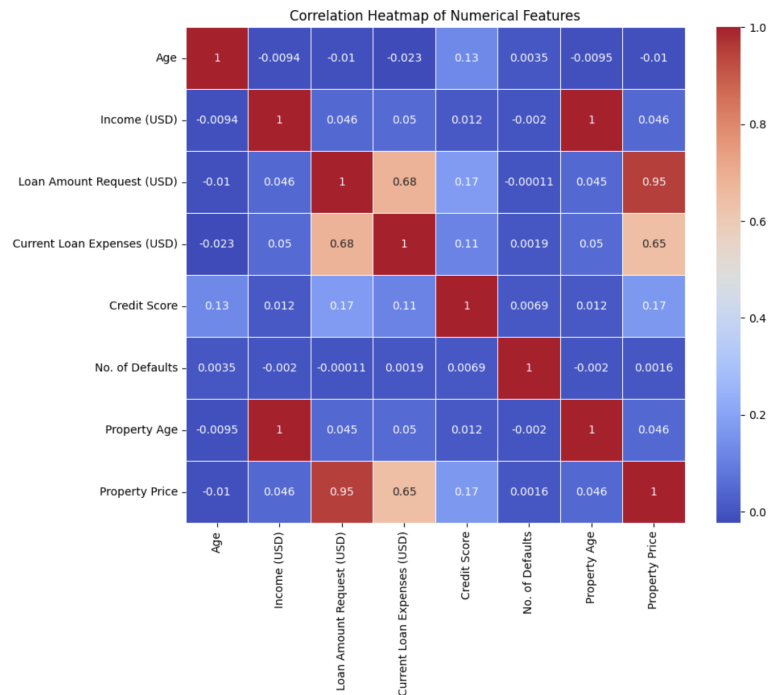


Figure 2: Correlation heatmap showing relationships between numerical variables.

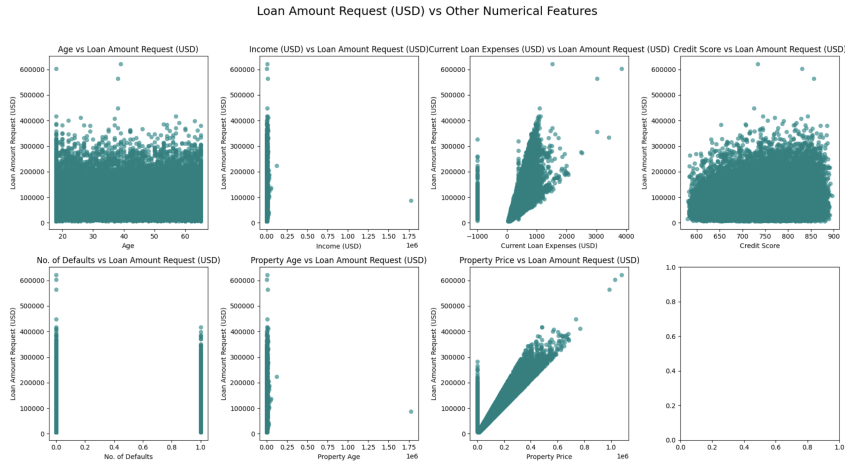


Figure 3: Scatter plot of applicant Income vs. Loan Amount Request.

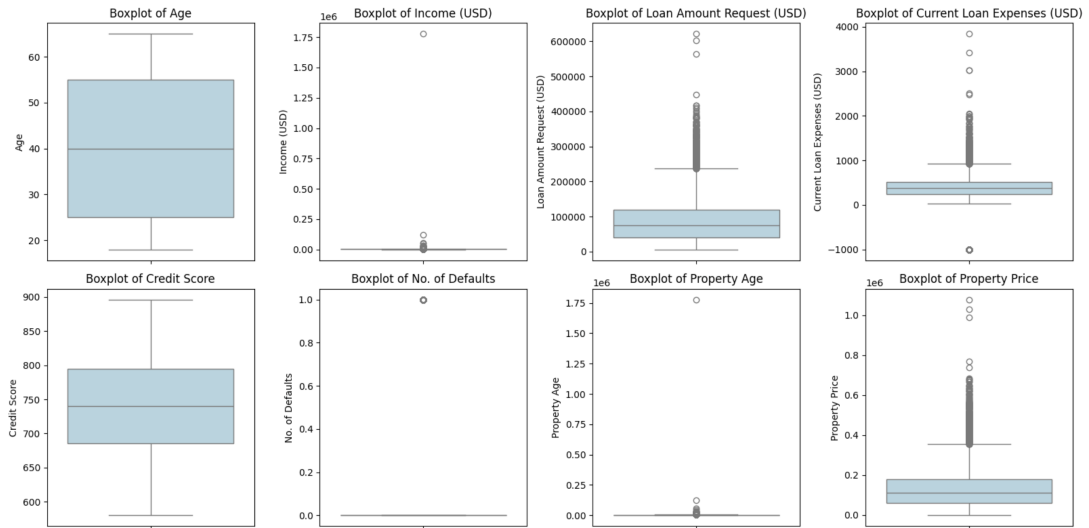


Figure 4: Boxplots of numerical features to identify outliers and distribution.

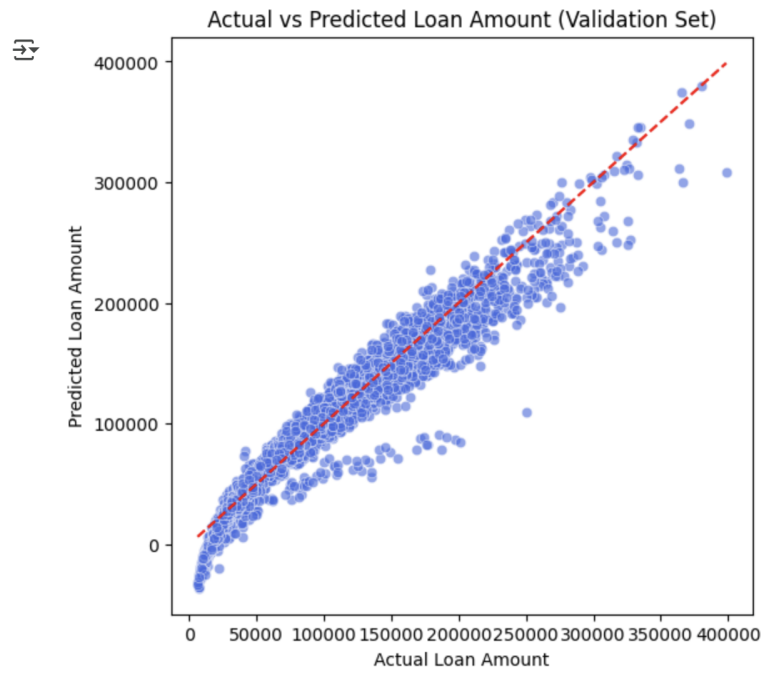


Figure 5: Actual vs. Predicted Loan Amount on the test set.

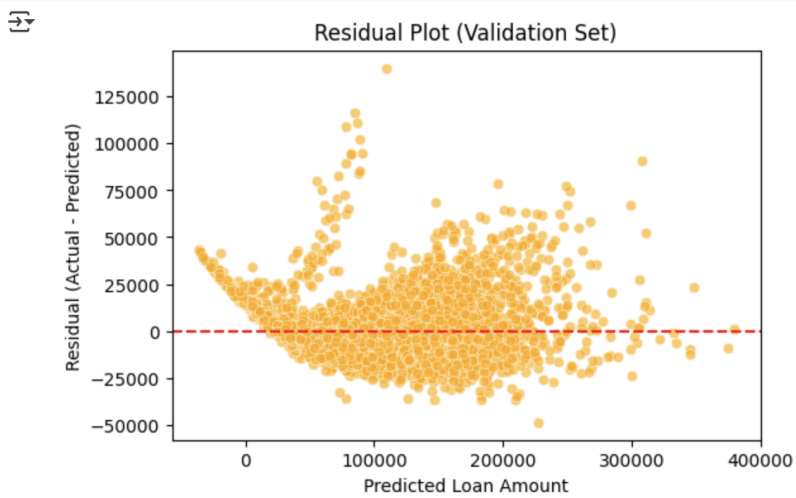


Figure 6: Residual plot to check the assumption of homoscedasticity.

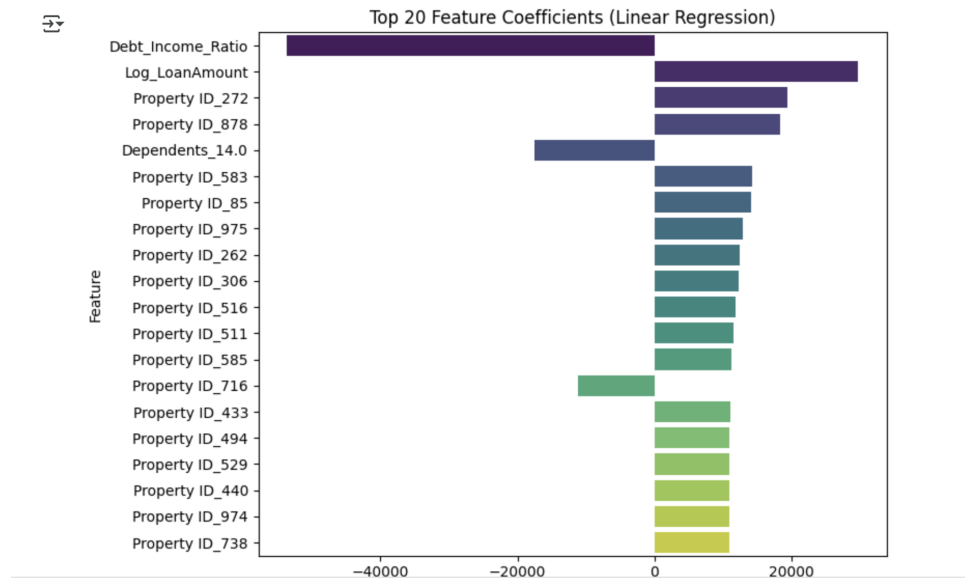


Figure 7: Top 20 most influential feature coefficients from the Linear Regression model.

7. Results

Table 1: Cross-Validation Results ($K = 5$)

Fold	MAE	MSE	RMSE	R ² Score
Fold 1	9970.74	213,160,706.45	14,600.02	0.9407
Fold 2	10032.56	208,914,681.38	14,453.88	0.9389
Fold 3	10293.25	933,662,717.80	30,555.89	0.7338
Fold 4	9729.49	189,575,514.40	13,768.64	0.9468
Fold 5	10023.68	204,815,861.41	14,311.39	0.9431
Average	10009.94	350,025,896.29	17,537.97	0.9007

Table 2: Summary of Results for Loan Amount Prediction

Description	Student's Result
Dataset Size (after preprocessing)	614 rows
Train/Test Split Ratio	80:20
Feature(s) Used for Prediction	All features after feature engineering and one-hot encoding
Model Used	Linear Regression
Cross-Validation Used? (Yes/No)	Yes
If Yes, Number of Folds (K)	5
Reference to CV Results Table	Table 1
Mean Absolute Error (MAE) on Test Set	10006.04
Mean Squared Error (MSE) on Test Set	214,528,533.92
Root Mean Squared Error (RMSE) on Test Set	14,646.79
R ² Score on Test Set	0.9403
Adjusted R ² Score on Test Set	Not calculated in the script
Most Influential Feature(s)	Features with the largest absolute coefficient values (e.g., Debt.Income.Ratio)
Observations from Residual Plot	Residuals show a pattern, suggesting the model's errors are not random.
Interpretation of Predicted vs Actual Plot	Predictions generally follow the trend of the actual values but with some variance.
Any Overfitting or Underfitting Observed?	No significant overfitting observed.
If Yes, Brief Justification	Performance on the validation and test sets are similar.

8. Best Practices

- Always check for and handle null values and potential outliers before model training.
- Use feature scaling (like StandardScaler) when features are on different scales to help the model converge effectively.
- Apply cross-validation to get a more robust estimate of the model's performance on unseen data.

- Visualize results, especially residual plots and actual vs. predicted plots, to validate the assumptions of Linear Regression.

9. Learning Outcomes

- Understood the end-to-end workflow of implementing a Linear Regression model, from data cleaning to evaluation.
- Gained practical experience in data preprocessing, EDA, feature engineering, and model evaluation techniques.
- Learned to correctly interpret key regression metrics, including MAE, MSE, RMSE, and the R^2 score.